Maulik Madhavi

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Experience

NCS Pte Ltd, Singapore, Senior AI Engineer — Apr 2021 – present, 5 years 7 months

- Contributed to KAI Create and KAI SnS, building and optimizing computer-vision and backend components for low-code video analytics and search/summarization workflows.
- Designed and implemented a million-scale face indexing system and person/vehicle detection pipelines with attribute recognition.
- Built a prompt studio for efficient video search, summarization, and analysis, working across product, engineering, and research teams.
- Fine-tuned open-weight LLMs (Mistral-7B, Llama-2-7B) and VLMs (Qwen3-VL, Moondream) with LoRA/SFT on internal Alpaca-style instruction datasets.
- Optimized vLLM serving for high-throughput video description generation, producing summaries for 48 concurrent 10-second chunks in ~7 seconds on a single H100 GPU.
- Optimized computer-vision models for edge and on-prem deployment through quantization, pruning, and distillation using ONNX Runtime, TensorRT, and OpenVINO.
- Developed an agentic AI workflow for autonomous computer-vision model training across planning, data collection, annotation, training, and evaluation.

National University of Singapore (NUS), Singapore, Research Fellow — Dec 2017 – Apr 2021, 3 years 5 months

- Designed and deployed a backend dialogue system for autonomous vehicles supporting multimodal conversations (FAQ, intent-driven, and open-domain) using Google Search API and Wikipedia.
- Developed a text-dependent speaker verification system optimized for mobile deployment.
- Improved ASR for healthcare by adapting language models to Singaporean English, yielding significant WER reductions.
- Built an automated form-filling system for SCDF call centers using NLU to extract actionable information from ASR transcripts.
- Mentored and co-supervised undergraduate and graduate students on research projects.

Dhirubhai Ambani Institute of Information and Communication Technology (DA-IICT), Gandhinagar, Project Personnel — Apr 2012 – June 2014, 2 years 3 months

- Prosodically guided phonetic engine development for Gujarati and Marathi languages (Advisor: Prof. Hemant Patil)
- Developed a phonetic engine for IPA transcription, syllabification, and prosodic analysis across read, spontaneous, and lecture speech from multiple dialectal regions.
- Designed the end-to-end phonetic engine workflow using HMM-based phoneme recognition, spectral analysis for syllable segmentation, and source-characteristic analysis for suprasegmental and acoustic modeling.
- Built an audio search pipeline leveraging the phonetic engine for spoken-content retrieval and discovery in Indian language processing.
- Explored query-by-audio alignment techniques to improve representation and matching for spoken-content retrieval, laying the foundation for Ph.D. research.

Education

Ph.D. **Dhirubhai Ambani Institute of Information and Communication Technology (DA-IICT), Gandhinagar**, Information and Communication Technology (ICT) — July 2011 – Nov 2017
- Ph.D. in Information and Communication Technology (ICT)
- CPI: 9.75/10
- Thesis: Design of QbE-STD System: Audio Representation and Matching Perspective

M.Tech **Dhirubhai Ambani Institute of Information and Communication Technology (DA-IICT), Gandhinagar**, Information and Communication Technology (ICT) — July 2009 – June 2011
- Master of Technology (M.Tech) in Information and Communication Technology (ICT)
- Communication Systems specialization
- CPI: 8.26/10
- Thesis: Person Recognition from Their Hum

B.E. **V.V.P. Engineering College, Saurashtra University, Rajkot**, Electronics and Communication (EC) — July 2005 – June 2009
- Bachelor of Engineering (B.E.) in Electronics and Communication (EC)
- Percentage: 69.73% (aggregate)

Publications

Vocal Tract Length Normalization using a Gaussian mixture model framework for query-by-example spoken term detection — Nov 2019
M. C. Madhavi, H. A. Patil
10.1016/j.csl.2019.03.005 (Computer, Speech and Language, Elsevier, vol. 58, pp. 175-202)

Design of mixture of GMMs for query-by-example spoken term detection — Nov 2018
M. C. Madhavi, H. A. Patil
10.1016/j.csl.2018.04.006 (Computer, Speech and Language, Elsevier, vol. 52, pp. 41-55)

Combining evidences from magnitude and phase information using VTEO for person recognition using humming — Nov 2018
H. A. Patil, M. C. Madhavi
10.1016/j.csl.2018.04.006 (Computer Speech and Language, Elsevier, vol. 52, pp. 225-256)

Partial matching and search space reduction for QbE-STD — Sep 2017
M. C. Madhavi, H. A. Patil
10.1016/j.csl.2018.04.006 (Computer, Speech & Language, Elsevier, vol. 45, pp. 58-82)

Static and dynamic information derived from source and system features for person recognition from humming — Sep 2012
H. A. Patil, M. C. Madhavi, K. K. Parhi
10.1007/s10772-012-9161-5 (Int. Journal of Speech Technology, Springer, vol. 15, no. 3, pp. 393-406)

Knowledge Distillation from BERT Transformer to Speech Transformer for Intent Classification — Aug-Sep 2021
Y. Jiang, B. Sharma, M. Madhavi, H. Li
10.21437/Interspeech.2021-402 (Interspeech, Brno, Czech Republic, pp. 4713-4717)

Diagnosis of COVID-19 using Auditory Acoustic Cues — Aug-Sep 2021
R. Das, M. Madhavi, H. Li
10.21437/Interspeech.2021-497 (Interspeech, Brno, Czech Republic, pp. 921-925)

Speaker-Utterance Dual Attention for Speaker and Utterance Verification Oct 2020
T. Liu, R. Das, M. Madhavi, S. Shen, H. Li
10.21437/Interspeech.2020-2656 (Interspeech, Shanghai, China, pp. 4293-4297)

Multi-target DoA estimation with an audio-visual fusion mechanism Jun 2021
X. Qian, M. Madhavi, Z. Pan, J. Wang, H. Li
10.1109/ICASSP39728.2021.9413776 (IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP), Toronto, Canada, pp. 4280-4284)

Leveraging Acoustic and Linguistic Embeddings from Pretrained Speech and Language Models for Intent Classification Jun 2021
B. Sharma, M. Madhavi, H. Li
10.1109/ICASSP46034.2021.9413388 (IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP), Toronto, Canada, pp. 7498-7502)

Awards

IAPR Travel Scholarship: International Association for Pattern Recognition Travel Scholarship for presenting a paper at the International Conference on Biometrics (ICB 2012), Delhi, India. Paper: Significance of Magnitude and Phase Information via VTEO for Humming-Based Biometrics. (Grant reference: IAPR009)

GATE 2009 Qualification: Qualified in Electronics and Communication Engineering with All India Rank 2142

Invited Talks

Doctoral Consortium Talk: Design of QbE-STD System: Audio Representation and Matching Perspective at Summer School on Speech Signal Processing, DA-IICT, Gandhinagar

Overview of Query-by-Example Spoken Term Detection: Talk at ISCA supported Summer School on Advances in Speech and Audio Processing (ASAP), DA-IICT, Gandhinagar

Query-by-Example Spoken Term Detection - A template matching approach: 5-minute Ph.D. competition at the ISCA-supported Summer School on Speech Signal Processing (S4P), DA-IICT, Gandhinagar

Professional Activities

Conference Reviewer: INTERSPEECH (2018-2026), ICASSP (2021, 2023-2026), APSIPA 2018, IALP (2019-2023, 2025-26), ISCSLP 2024, Computer Speech and Language, Speech Communication (Elsevier), IEEE Access

Technologies

Research Areas: Speech signal processing, voice search, speaker verification, spoken language understanding, deep learning for spoken language processing, million-scale indexing

Programming Languages: Bash, C/C++, Python, JavaScript, MATLAB

Deep Learning & ML Frameworks: TensorFlow, PyTorch, OpenCV, scikit-learn, Hugging Face Transformers, NumPy, Pandas, Matplotlib

Large Language Models & AI Tools: LangChain, LlamaIndex, DSPy, vLLM, RASA, n8n, Agentic Harness (Claude Code, OpenCode, Copilot, make_my_harness)

LLM/VLM Fine-tuning & Serving: LoRA/SFT fine-tuning (Mistral-7B, Llama-2-7B), vision-language model tuning (Gemma, Moondream, BLIP, CLIP), high-throughput vLLM serving (PagedAttention, continuous batching, KV cache)

Model Optimization & Edge Deployment: Quantization (INT8/INT4), pruning, knowledge distillation, ONNX Runtime, TensorRT, OpenVINO, model export (ONNX/TorchScript)

Web & API Frameworks: Flask, FastAPI, Streamlit, Gradio

Audio/Speech Tools & Systems: Kaldi, wakeup-word detection, voice activity detection

DevOps & Deployment: Git, Docker, Kubernetes

Operating Systems: Windows and Linux